

ABOUT BAWARCHIKHANA

BawarchiKhana (باورچی خانہ) is an AI-powered cooking assistant platform designed specifically for Pakistan and the South Asian market. The name means "kitchen" in Urdu — and that's exactly what it is: your intelligent digital kitchen companion.

WHAT IS IT?

BawarchiKhana is a full-stack web application that combines the power of Google's Gemini AI with a deep understanding of Pakistani cuisine, halal dietary requirements, and local cooking culture. It helps everyday home cooks make smarter decisions in the kitchen — from what to cook tonight based on what's already in their pantry, to generating full weekly meal plans, to getting step-by-step AI-guided cooking instructions.

THE PROBLEM IT SOLVES

Pakistani home cooks face several everyday challenges:

1. Food Waste — Ingredients bought for one recipe go unused and expire. With BawarchiKhana, users input what they have and get recipes that use exactly those ingredients.
2. Recipe Discovery — Most AI cooking tools are built for Western cuisines. Pakistani cooks need a tool that understands biryani, nihari, daal, and the complexity of regional spice blends.
3. Halal Confidence — Knowing whether a recipe or ingredient substitution is halal requires domain knowledge. BawarchiKhana builds this in from day one.
4. Language Accessibility — Urdu voice support makes the platform accessible to users who are more comfortable in their native language.
5. Meal Planning — Deciding what to cook across a whole week is mentally exhausting. AI-powered meal planning takes that burden away.

HOW IT WORKS

Users sign up with their phone number (OTP via SMS) or Google account. They set up their pantry by listing what ingredients they have at home. From there, BawarchiKhana's AI engine —

powered by Google Gemini — generates personalized recipe suggestions, builds weekly meal plans, identifies food from photos, and acts as an interactive copilot while cooking.

Users can export their meal plans and recipes directly to Google Sheets and Google Drive, making it easy to shop, plan, and share.

For users who want more, a premium subscription (paid via JazzCash or EasyPaisa — Pakistan's most popular mobile payment platforms) unlocks advanced AI features, cooking masterclasses, and exclusive recipe content.

TECHNOLOGY

BawarchiKhana is built on a modern, production-grade tech stack:

- Frontend: Next.js (React) deployed on Vercel for fast global access
- Backend: NestJS (Node.js) deployed on Google Cloud Run — serverless, scalable, and always available
- Database: PostgreSQL on Google Cloud SQL with Prisma ORM for type-safe data access
- AI Engine: Google Gemini AI for all intelligent features
- Authentication: JWT tokens + Google OAuth2 + Twilio OTP
- Payments: JazzCash and EasyPaisa gateway integrations
- Infrastructure: Fully containerized (Docker) with CI/CD via Google Cloud Build

The backend exposes a clean REST API with modules for authentication, AI features, recipe management, user profiles, payments, and Google Drive integration. The entire deployment pipeline — from code push to live production — runs on Google Cloud Platform in the Asia-South1 (Mumbai) region, keeping latency low for Pakistani users.

WHO IS IT FOR?

- Home cooks in Pakistan who want smart recipe guidance without wasting ingredients
- Young professionals who need quick, halal meal ideas during busy weekdays
- Families who want to plan their week's meals efficiently
- Food enthusiasts who want to explore regional Pakistani cuisine with AI assistance

CURRENT STATUS

BawarchiKhana is live and deployed:

- Frontend: <https://bawarchi-khana-pi.vercel.app>
- Backend API: <https://bawarchikhana-backend-248619180591.asia-south1.run.app>

The platform is under active development. Core features including AI recipe suggestions, meal planning, pantry management, cooking sessions, Google Drive integration, and payment processing are fully functional.

WHAT'S NEXT

The roadmap includes Urdu voice synthesis for hands-free cooking guidance, a mobile app for iOS and Android, community features for recipe sharing, automatic pantry updates via receipt scanning, and a full Redis caching layer for improved performance.

Built by Fuzail Faraz — a developer passionate about bringing modern AI technology to everyday Pakistani life.

GitHub: <https://github.com/fuzailfaraz/BawarchiKhana>